Description of the laboratory work No. 5

SPI PROGRAMMING OF 32-BIT MICROCONTROLLERS (STM32F429I-DISC1)

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1. Objectives

Get to know the architecture, functionality and configuration features of SPI serial interfaces on the STM32F429I-DISC1 board. Develop the ability to analyze the microcontroller and board resources and to build simple loopback and master-slave SPI demonstrations.

2. Preparation for the work

We will work with the SPI serial interfaces of the STM32F429I-DISC1 board during this laboratory work. Because this board has only two user LEDs, the received SPI data will be

demonstrated mainly on the onboard TFT display, while LEDs PG13 and PG14 can be used as transfer activity indicators.

3. Workflow

3.1. Work with a teacher

The lecturer demonstrates the STM32F429I-DISC1 board, explains the available SPI resources and the jumper connections, and then students start to program individually.

3.1.1. STM32CubeIDE

The detailed project configuration has already been discussed in the previous laboratory works, therefore here only the essential configuration windows for laboratory work No. 5 are important. In the adapted STM32F429I-DISC1 project the LCD-related peripherals (LTDC, FMC, SPI5 and I2C3) are already prepared, while students must configure SPI3, SPI4 and the timers used in their own variant.

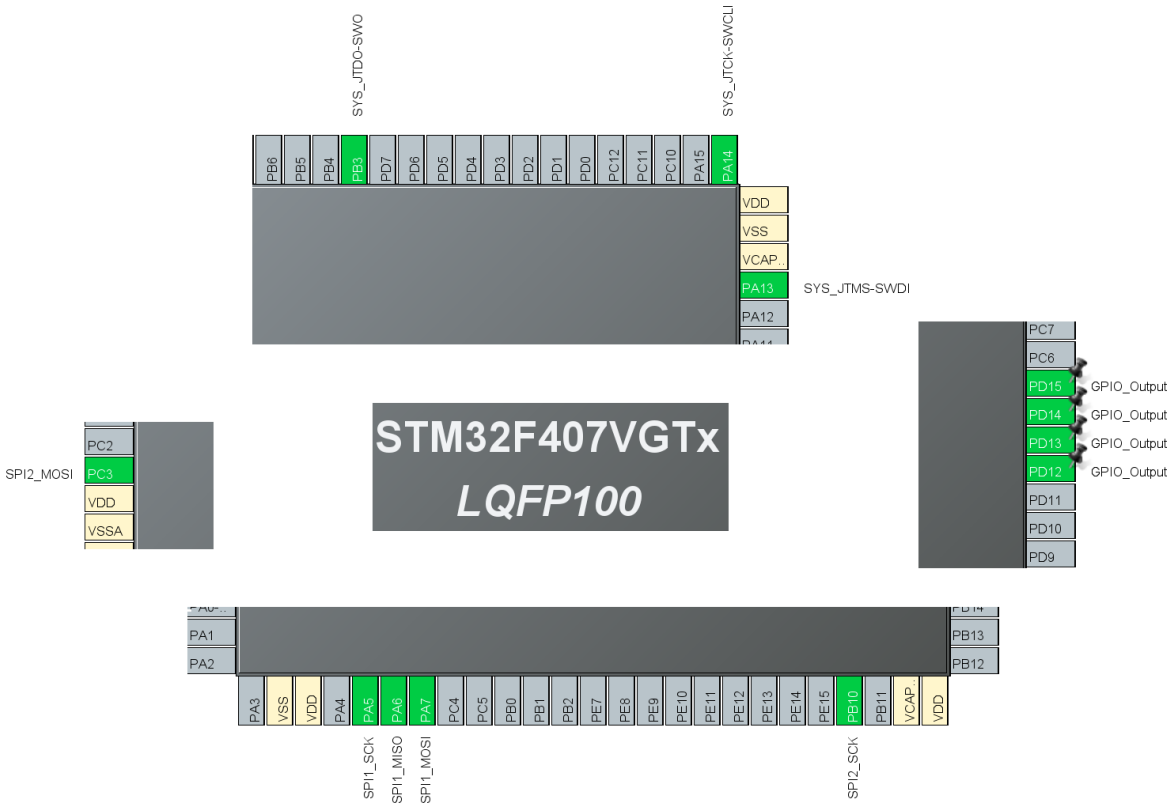


Fig. 1. SPI3, SPI4 and PG13/PG14 activity LED configuration for the STM32F429I-DISC1 board. The transferred values are displayed on the onboard TFT screen.

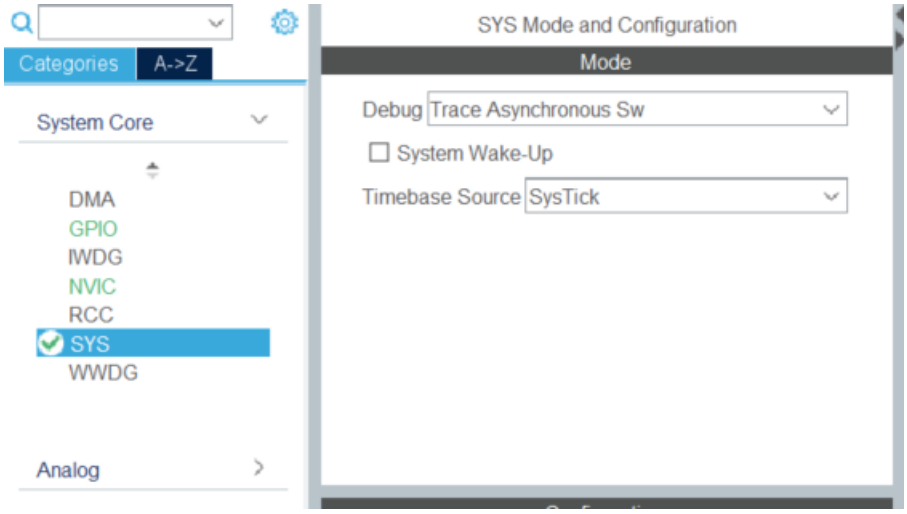


Fig. 2. If SPI3 uses PB3 as SCK, leave Trace Asynchronous SWO disabled so that PB3 can be used as a regular SPI pin.

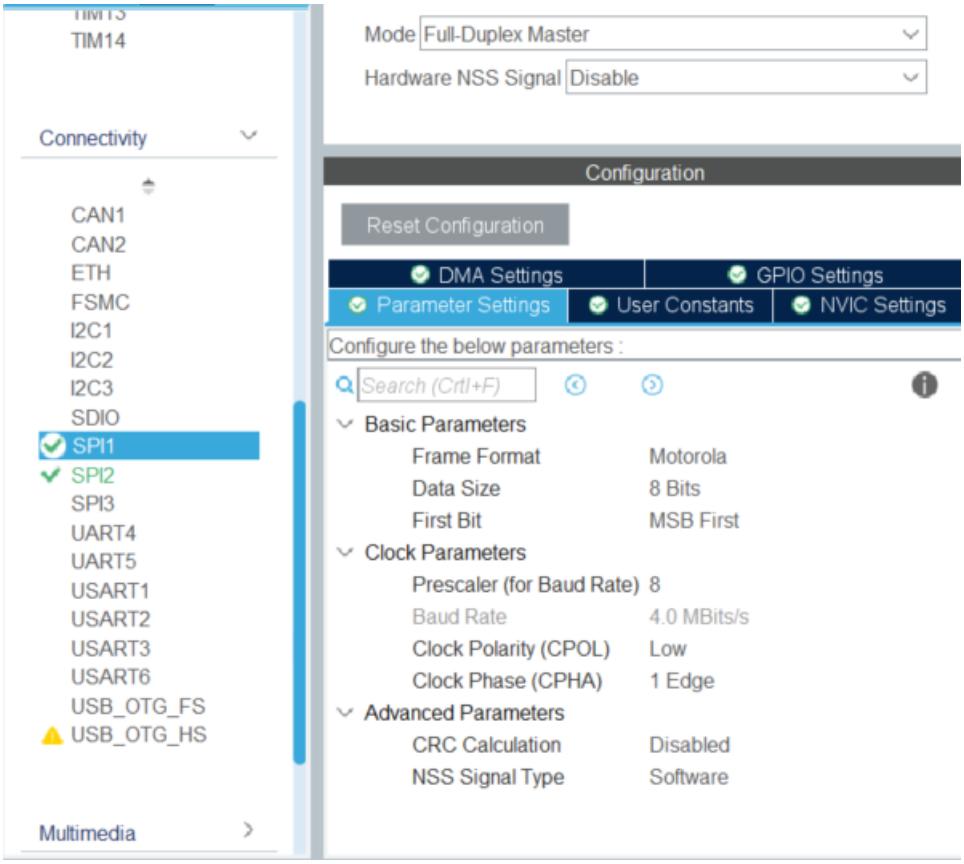


Fig. 3. SPI4 configuration when the interface should work in Master Full Duplex mode.

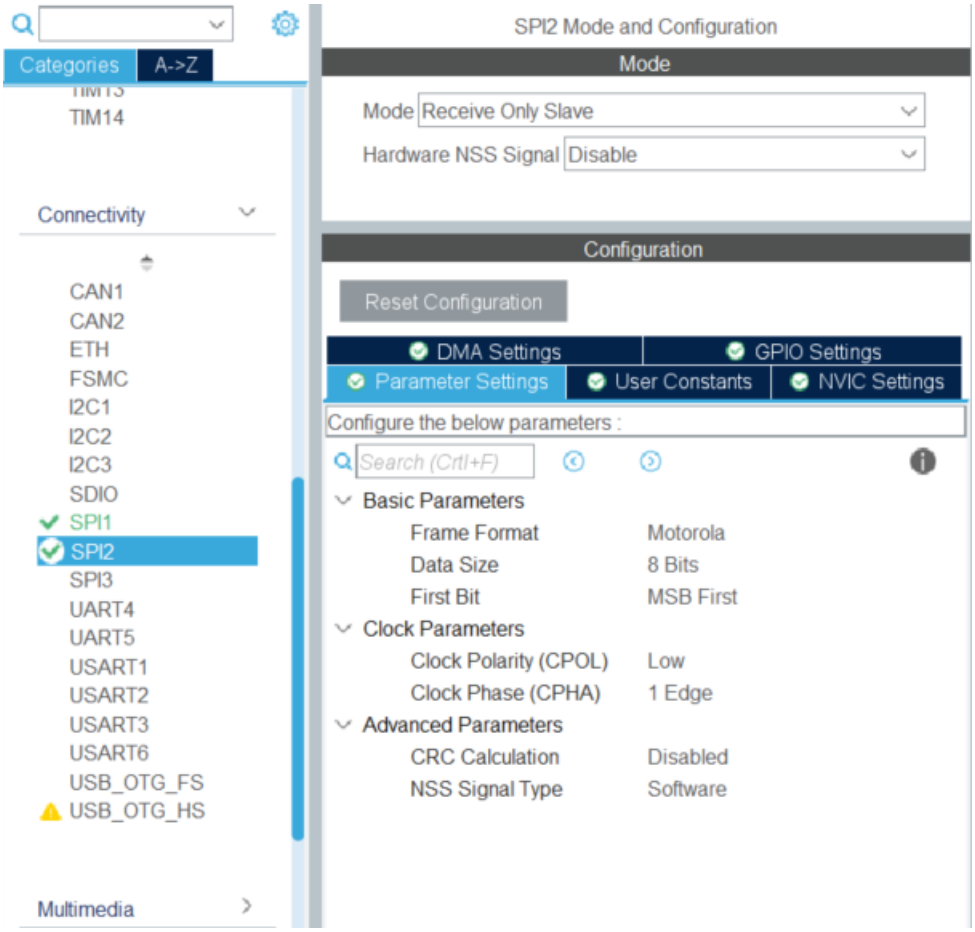


Fig. 4. SPI3 or SPI4 configuration when the interface should work in Slave Receive Only mode.

Note. These are generalized examples of configuration. In every specific variant it is necessary to adjust the SPI instance, the timer instance and the TFT output text according to the selected task.

3.1.2. Template of the code

Examples of code templates are provided in the descriptions of previous laboratory works. In this adapted version it is practical to work with main.c, stm32f4xx_it.c and a separate application file that initializes SPI3/SPI4, TIM2/TIM3 and the TFT output.

3.1.3. Main functions in laboratory work No. 5

- HAL_SPI_TransmitReceive(…) – blocking mode: SPI transmit and receive for the mandatory loopback part.
- HAL_TIM_Base_Start_IT(…) – non-blocking start of TIM2/TIM3 for 1 s periodic transfers.
- HAL_SPI_Receive_IT(…) – arm the slave SPI peripheral for non-blocking byte reception.
- HAL_SPI_Transmit(…) – transmit one byte from the master SPI in the timer driven scenario.
- HAL_TIM_PeriodElapsedCallback(…) / TIMx_IRQHandler(…) – generate the 1 s transfer event and handle the timer interrupt.
- HAL_SPI_RxCpltCallback(…) / BSP_LCD_DisplayStringAt(…) – process the received byte and update the onboard TFT display.

Data structures:

- uint8_t array[2] = {1, 2}; – two cell array.

- int a =10; a – value 10 will be returned; &a – address of the value 10 in the memory will be returned.

3.2. Work by individual task

Students independently perform the individual work assignments provided by the lecturer. They must connect the required jumper wires, run their own SPI configuration and demonstrate the transferred values on the TFT screen. When faced with difficulties, students can ask specific questions to the teacher.

3.3. Literature

1. Slides of the lectures in the moodle.vgtu.lt system.
2. STM32CubeIDE / STM32CubeMX code examples and the provided LAB5 project template.
3. STM32F429xx reference manual and datasheet.
4. STM32F429I-DISC1 user manual.
5. STM32F429I-DISC1 board schematic and pinout information.
6. SPI and LTDC/BSP LCD examples from the STM32CubeF4 firmware package.
7. Materials from previous laboratory works and the adapted LAB5 code example.